

Multimodal Deep Learning for Pulmonary Embolism Prognosis Prediction

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Abstract

Pulmonary embolism (PE) represents a critical medical condition characterized by the sudden obstruction of the pulmonary artery by blood clots, necessitating prompt intervention. Despite advancements in clinical methods utilizing pixel-centric deep learning models, valuable insights stored in Electronic Health Records (EHRs) often remain untapped. When applying deep learning to classify PE in computed tomography pulmonary angiography (CTPA), significant challenges emerge due to the complexity of CTPA examinations. To address this, integrating clinical and imaging data through multimodal approaches becomes imperative. Leveraging the "RadFusion Dataset," our study explores the efficacy of combining TabNet for clinical data and PENet for imaging data, aiming to surpass existing benchmarks. Advanced fusion techniques are employed to enhance prediction accuracy and interpretability. Feature importance analysis is conducted to develop a real-time prototype for clinical decision support. This project seeks to refine PE prognosis accuracy, bridging the gap between deep learning models and clinical practices, thereby advancing AI applications in medical research and consultations.

Index Terms

Pulmonary Embolism, Deep Learning, Multimodal fusion, Clinical Decision Support

I. INTRODUCTION

Pulmonary embolism (PE) poses a critical threat, marked by the sudden blockage of pulmonary arteries, often initiated by blood clots originating in the legs or other body parts. This obstruction causes severe respiratory and cardiovascular consequences, necessitating a strategic and timely intervention for effective management. A key aspect of this strategy lies in risk stratification, a guiding principle influencing diagnostic and therapeutic choices. In routine clinical scenarios, healthcare practitioners heavily rely on the patient's electronic health record (EHR) to contextualize their interpretation of medical imaging findings. The common practice entails a consideration of either clinical or imaging data. However, prevalent deep learning models in radiology often focus narrowly on pixel-value information, disregarding valuable insights within the broader health record.

Despite the potential demonstrated by deep learning in various medical imaging tasks, applying it to automated pulmonary embolism (PE) classification in computed tomography pulmonary angiography (CTPA) studies presents unique challenges. CTPA examinations, with their larger scale compared to conventional medical imaging tests, demand a more delicate approach. The pixel data related to PE findings represents only a fraction of the extensive 3D CTPA volume. Thus, the fusion of patient EHR data with imaging data emerges as a crucial aspect for efficiency. This introduces the concept of multimodal approaches in medical research.

The integration of multimodal approaches, such as the fusion of clinical and imaging data in PE prediction models, holds immense significance. These models offer a more comprehensive understanding of patient conditions by integrating diverse information sources, thereby enhancing diagnostic accuracy and informing tailored treatment strategies. However, the limited availability of publicly accessible datasets poses a challenge in developing effective PE prediction models. Also most of the prevalent deep learning models are trained on smaller datasets.

The paper "RadFusion" addresses this gap by introducing a multimodal benchmark dataset for pulmonary embolism detection, combining 3D medical imaging with EHR data of 1794 patients. This dataset, by our knowledge is so far the largest multimodal that is publically available. They proposed the idea of exploration of multimodal fusion for the prognosis of pulmonary embolism while the researches are being done for the other popular diseases. Their findings suggest that, rather than focusing solely on advancing medical image representation models, there should be an increased emphasis on enhancing research in the creation of improved multimodal fusion models. However on a little research, most of the other multimodal fusion models which shows greater performances are often trained and tested on smaller datasets as the limited availability of this confidential medical data is the greatest barrier to the researchers in the area.

This sets the stage for our project, which aims to address specific challenges in pulmonary embolism prognosis through an exploration of one among the best deep learning models in their respective area of performance—particularly TabNet and PENet. Our goal is to extract complicated features from both clinical and medical imaging data, uncovering undiscovered patterns crucial for prognosis. Leveraging the strengths of TabNet and PENet, our proposed deep learning model utilises fusion mechanisms, enhancing overall predictive capability. Our work aims to build an efficient deep learning, evaluating and surpassing the benchmark set by the RadFusion dataset.

In our attempt, we not only seek to develop a robust multimodal fusion model for efficient PE prognosis prediction but also aim to create a prototype tool which can support clinical decision making. This tool harnesses the power of an NLP-based integration using the help of LLM tools to empower healthcare professionals in seamlessly integrating EHR data into the predictive framework. The proposed interface allows doctors to input the EHR report, upload relevant CTPA data, and initiate the model.

II. RELATED WORKS

Previous research in pulmonary embolism (PE) detection has primarily focused on utilizing either imaging data or clinical records in isolation, leading to limitations in diagnostic accuracy and interpretability. In response to this, recent studies have explored the integration of Electronic Health Records (EHRs) with imaging data through multimodal fusion techniques.

Several previous studies have explored the application of deep learning techniques for pulmonary embolism (PE) detection using various imaging modalities such as computed tomography pulmonary angiography (CTPA). Tajbakhsh et al. [1] demonstrated the efficacy of 2D convolutional neural networks in PE detection. Subsequent studies have employed various methods ranging from 3D convolutional neural networks to architectures built upon ResNet and DenseNet. Recent advancements in deep learning architectures have led to the development of specialized models for analyzing medical imaging data, offering improved accuracy and efficiency. Huang et al. [5] proposed PENet, a state-of-the-art deep learning model optimized for pulmonary embolism detection in CTPA images, showcasing superior performance compared to traditional approaches.

Multimodal fusion techniques have emerged as a promising strategy to leverage complementary information from diverse data sources, enhancing the robustness and interpretability of predictive models. Chen et al. [9] employed deep learning and machine learning for type 2 diabetes risk prediction with diverse modalities. Zhou et al. [10] proposed a multilayer framework for Alzheimer’s detection and progression using multiple modalities (imaging, clinical, etc.) with a strong focus on explainability. The significance of the fusion of Electronic Health Data along with Imaging CTPA data was addressed by Zhou et al. [7].

Tabular data processing techniques have been widely employed in medical diagnosis tasks, particularly for handling structured clinical data extracted from EHRs. Arik et al. [11] introduced TabNet, a novel deep learning architecture tailored for tabular data, demonstrating its effectiveness in disease prediction tasks with high-dimensional clinical data. As the dataset available from the RadFusion study is presented in a CSV file format, our aim was to improve the predictions using a model that gives state-of-the-art performance in processing tabular data.

Ensemble learning methods have been employed for decision-making and classification tasks. Mienye and Sun [14] conducted a survey on ensemble learning, providing insights into concepts, algorithms, applications, and prospects. In our study, we employed decision-level late fusion strategy using stacking, where several meta learner models were employed and the one which gave the most accuracy was chosen.

III. METHODOLOGY

In this methodology, we present a comprehensive approach for developing a multimodal fusion model aimed at pulmonary embolism (PE) classification, integrating information from both imaging data and electronic health records (EHR).

A. Dataset

The dataset utilized in this study was obtained from the RadFusion database [7], a comprehensive collection of medical data curated for research purposes, focusing on pulmonary embolism (PE) detection. Approval for dataset access was obtained from Stanford University Medical Center (SUMC) with the necessary ethical clearances from the Stanford Institutional Review Board (IRB). The RadFusion dataset includes both imaging data from CT scans and Electronic Health Records (EHRs) of patients. The dataset provides detailed information on patient demographics, vitals, medications, International Classification of Diseases (ICD) codes, lab test results, and annotations related to PE diagnosis. Annotations include the classification of PE subtypes and the identification of slices containing PE lesions. The dataset was then partitioned into training, validation, and testing sets in an 80% /10%/10% split, ensuring no patient overlap between the sets. The RadFusion dataset serves as a

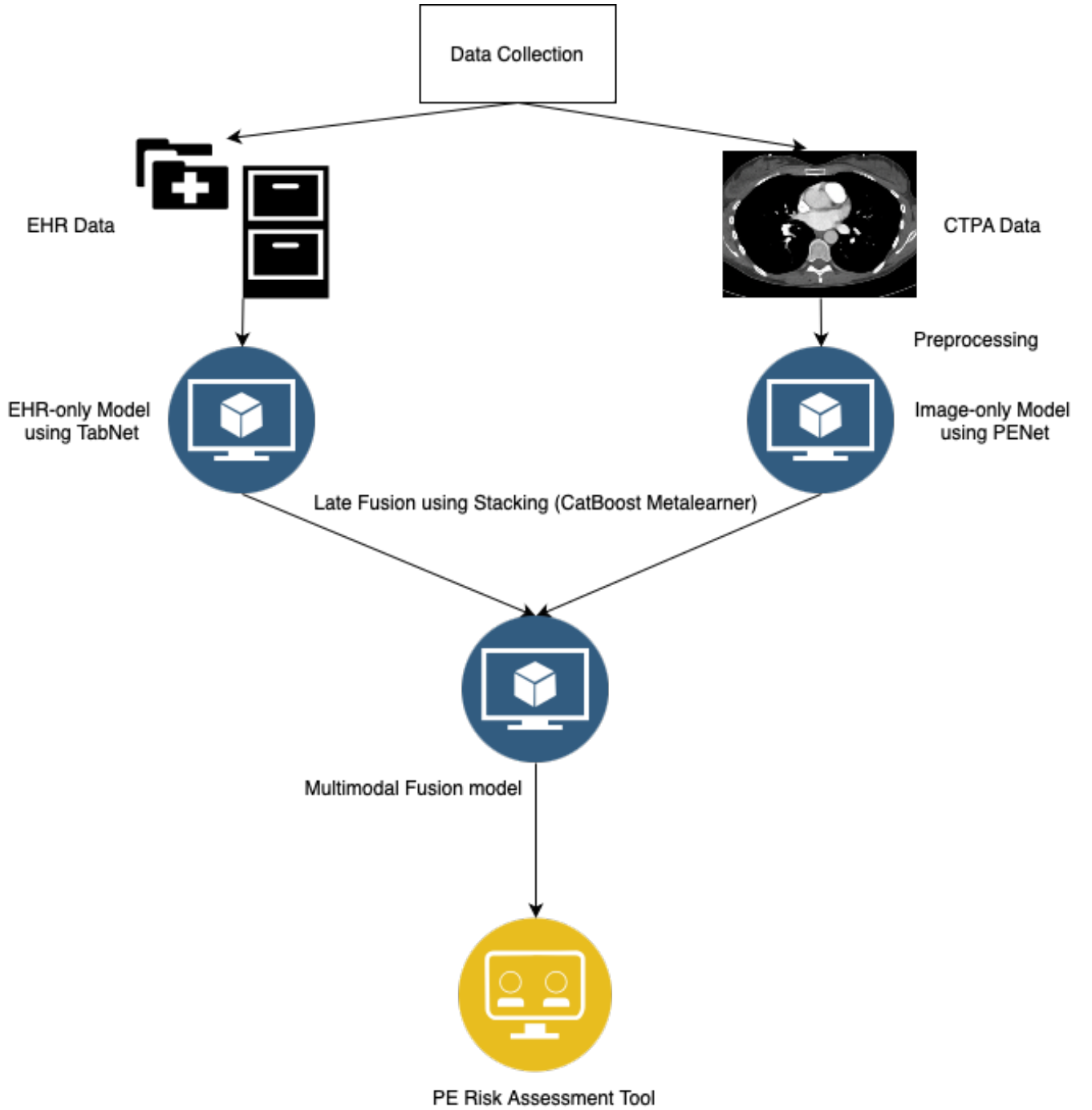


Fig. 1. Proposed Method

valuable resource for developing and evaluating machine learning algorithms for PE detection. It enables the exploration of multimodal fusion models using both CT scans and patient EHRs, contributing to advancements in medical image interpretation and clinical decision support systems. Researchers can utilize the dataset to study model fairness across different demographic groups and investigate algorithmic bias in medical AI applications.

1) Data Characteristics: Table I summarizes the data characteristics of the RadFusion dataset, providing statistics for the training, validation, and testing sets. These characteristics include the number of studies, patients, slices, and the distribution of positive and negative PE cases. Additionally, vital signs, D-dimer tests, and BMI information are provided, offering insights into patient health records. The dataset is enriched with patient demographics, enabling studies on model fairness and demographic-based analysis.

Category	Sub-category	Overall	Train	Validation	Test
CTPA exams	# of studies	1837	1454	193	190
	# of patients	1794	1414	190	190
	Median # of slices (IQR)	386 (134)	385 (136)	388 (132)	388 (139)
PE	# of negative PE	1111(60.48%)	946 (65.06%)	85 (44.04%)	80 (42.10%)
	# of positive PE	726 (39.52%)	508 (34.94%)	108 (55.96%)	110 (57.89%)
	Central	257(35.40%)	202 (39.76%)	27 (25.00%)	28 (25.45%)
	Segmental	387(53.31%)	281 (55.31%)	52 (48.15%)	54 (49.09%)
	Subsegmental	82 (11.29%)	25 (4.91%)	29 (26.85%)	28 (25.45%)
Vitals	BMI (mean: std)	28.37 : 9.65	28.36 : 10.03	27.11 : 6.78	29.60 : 9.22
	Pulse (mean: std)	81.62 : 14.99	81.53 : 15.64	83.05 : 11.86	80.50 : 13.06
D-dimer	D-dimer test taken	580 (30.62%)	461 (30.90%)	58 (28.71%)	61 (30.50%)
	D-dimer positive	496 (26.18%)	389 (26.07%)	51 (25.25%)	56 (28.00%)

TABLE I
DATA CHARACTERISTICS OF THE RADFUSION DATASET.

B. Image-only model

For the image-only model, we employ the PENet architecture, which stands among the forefront of 3D convolutional neural networks (CNNs) for pulmonary embolism (PE) detection. PENet is adept at processing volumetric computed tomography pulmonary angiography (CTPA) scans, leveraging spatial information across multiple slices to detect subtle abnormalities indicative of PE. The architecture comprises four key components: the PENet unit, Squeeze-and-Excitation (SE) block, PENet bottleneck, and PENet encoder.

The PENet unit serves as the fundamental building block, employing 3D convolutions, group normalization, and LeakyReLU activation to extract features from the input data. The SE block enhances feature recalibration by adaptively weighting channel-wise information, promoting discriminative feature learning. Furthermore, the PENet bottleneck aggregates multiple PENet units to form a hierarchical feature extractor, facilitating the representation of complex spatial patterns.

The PENet encoder, consisting of stacked PENet bottlenecks followed by GapLinear activation, culminates in the final prediction. Notably, the depth of the network is optimized through cross-validation, ensuring a balance between model complexity and generalization performance.

In preparation for model training, we preprocess the CTPA image data to enhance model efficacy. Each CT scan undergoes preprocessing steps, including resizing each slice to 224×224 pixels, and applying an optimized viewing window centered around pulmonary arteries (window center = 400, window width = 1000). Additionally, we clip Hounsfield Units to the range of -1000 to 900 and normalize each CT scan to be zero-centered. The training strategy incorporates techniques to address class imbalance, such as binary cross-entropy focal loss and up-sampling of positive windows. Data augmentation methods, including random cropping, rotation, and jittering, are employed to improve model robustness.

C. EHR Model

In developing the EHR model, we deploy the TabNet algorithm, a cutting-edge deep learning framework tailored explicitly for processing tabular data. Distinguished from conventional regression models, TabNet boasts a unique architecture that seamlessly integrates feature selection and decision-making mechanisms, making it particularly adept at managing high-dimensional and sparse input features characteristic of EHR datasets. By leveraging the regularization penalties of L1 and L2, TabNet strikes an optimal balance between model complexity and generalization capacity, thereby augmenting its proficiency in discerning meaningful patterns inherent in the data.

Furthermore, the EHR data from disparate CSV files are consolidated, followed by an initial training phase wherein a TabNet model is trained. Subsequently, leveraging the feature importance matrix, non-contributing columns are pruned, and the TabNet model is retrained. This iterative process optimizes the model's efficacy by focusing solely on salient features conducive to accurate predictions. Finally, the feature importance analysis is conducted to ascertain the significance of individual features in influencing model predictions, thereby offering valuable insights for clinical decision-making.

D. Multimodal Fusion

In this section, we detail the construction of our multimodal fusion model, which integrates predictions from individual models trained on both imaging data and Electronic Health Records (EHRs). Our primary fusion strategy revolves around late fusion, also known as decision-level fusion, wherein the predictions from individual models are combined using various meta-learning algorithms, including TabNet, XGBoost, and CatBoost. This approach enables us to leverage the complementary strengths of each modality while mitigating their respective limitations, thereby enhancing overall prediction performance.

Late fusion involves aggregating the predictions generated by individual models at the decision level, typically by averaging or applying a weighted combination of the model outputs. We experiment with several late fusion techniques to determine the most effective fusion strategy for our multimodal model. Additionally, we explore the use of ensemble learning methods, such as stacking, to further improve prediction accuracy and robustness.

Algorithm 1 outlines the late fusion process used in our multimodal model. Given the predictions from individual models for a given sample, the algorithm combines these predictions using a specified fusion method, such as simple averaging or weighted averaging based on model performance. The fused prediction is then used as the final output of the multimodal model.

Algorithm 1: Stacking Method for Multimodal Fusion

Input: Training datasets: D_{EHR} , D_{CTPA}

Output: Fused prediction: P_{fused}

- 1 Train EHR Model: Train the EHR model using the TabNet algorithm on the EHR training dataset D_{EHR} ;
 - 2 Train CTPA Model: Train the CTPA model using the PENet architecture on the CTPA training dataset D_{CTPA} ;
 - 3 Generate Predictions: Generate predictions P_{EHR} and P_{CTPA} from the trained EHR and CTPA models, respectively, for the validation/test datasets;
 - 4 Combine Predictions: Combine the predictions P_{EHR} and P_{CTPA} to create a combined feature matrix;
 - 5 Train Meta-Learner Model: Train a meta-learner model (e.g., XGBoost, CatBoost) on the combined feature matrix to learn the relationship between individual model predictions and the target variable;
 - 6 Generate Fused Prediction: Generate the fused prediction P_{fused} by feeding the predictions from the EHR and CTPA models into the trained meta-learner model;
 - 7 **Return:** P_{fused}
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Furthermore, we evaluate the performance of the multimodal fusion model using appropriate metrics, such as accuracy to assess its effectiveness in predicting pulmonary embolism. Through systematic experimentation and analysis, we aim to identify the optimal fusion strategy and meta-learning algorithm combination that maximizes prediction accuracy while ensuring robustness. The multimodal fusion model represents a significant advancement in leveraging heterogeneous data sources for medical diagnosis, offering potential benefits in enhancing diagnostic accuracy and clinical decision-making in the context of pulmonary embolism detection.

E. Clinical Decision Support Interface

In order to demonstrate the practical application of our multimodal fusion model for clinical decision support, we developed an interactive interface prototype. This interface serves as a tool for healthcare professionals to input patient data and obtain predictions for pulmonary embolism (PE) risk.

For the Electronic Health Record (EHR) input, we first identify the highest feature importance columns from the TabNet model. These columns represent the most influential factors in predicting PE risk based on the EHR data. We then design a set of question-answer-based inputs corresponding to these key features. Healthcare professionals can input patient information through this interface, answering questions related to vital signs, medical history, and other relevant factors.

The interface dynamically processes the user inputs and generates predictions using our multimodal fusion model, thereby eliminating the need to input all the features upon which the model is trained in. By leveraging both EHR and imaging data, our model provides a comprehensive assessment of PE risk for the given patient. The output is displayed to the user, indicating the predicted likelihood of PE occurrence along with any additional diagnostic insights.

This prototype interface demonstrates the potential of integrating advanced machine learning models into clinical practice for improved decision-making support. By leveraging patient data and state-of-the-art algorithms, healthcare professionals can make more informed decisions and enhance patient care outcomes.

IV. RESULT

A. Model Performance

The Multimodal Models integrate both image and EHR data for improved predictive performance. The CatBoost Fusion model outperformed other models with an accuracy of 0.93. It achieved a precision of 0.87 and a recall of 0.95. The F1 score for this model is 0.90, and the AUC is 0.96, indicating excellent discriminatory capability. Overall, the results indicate that the CatBoost Fusion model, leveraging multimodal data integration, achieves the highest performance across all evaluated metrics. This is in comparison with other decision level late fusion models using meta-learners of XGBoost, TabNet, and CatBoost models. Compared to the RAD Fusion benchmark, our multimodal fusion models demonstrated significant improvements in accuracy across all evaluated metrics. Specifically, surpassing the benchmark by 5 percentage points. This notable enhancement underscores the effectiveness of our late decision fusion approach in leveraging complementary information from different modalities to enhance predictive performance.

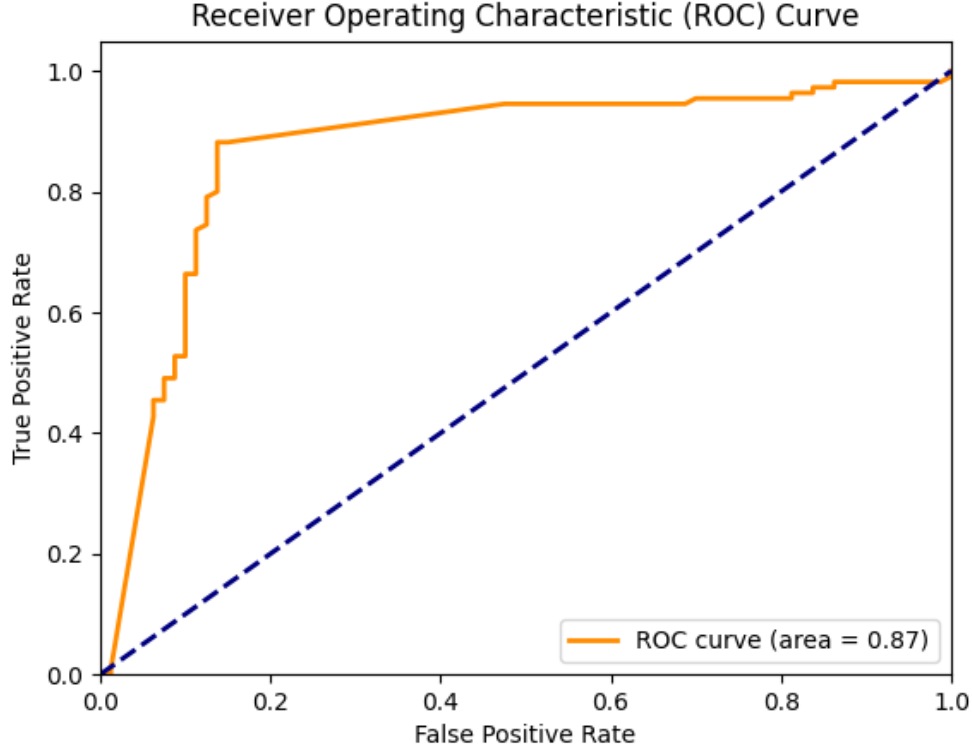


Fig. 2. Receiver Operating Characteristic (ROC) Curve

Figure 3 showcases the prediction trend for about 40 samples on how the prediction scores can be compared to their actual labels for the PENet model, the TabNet model, and the Multimodal model respectively. It can be understood that the predictions done by the multimodal model align more with the pattern of the actual labels.

B. Feature Importance

Our feature importance analysis revealed key insights into the predictive power of features extracted from the EHR data. Notably, diseases of pulmonary circulation emerged as the most influential feature, with a feature importance score of 0.041. This highlights the critical role of pulmonary health indicators in disease prediction and underscores the importance of monitoring and managing pulmonary-related conditions in clinical settings.

Furthermore, local anesthetics and topical antifungals were identified as important features, with feature importance scores of 0.015 and 0.010, respectively. These findings suggest that medications and treatments related to specific medical conditions

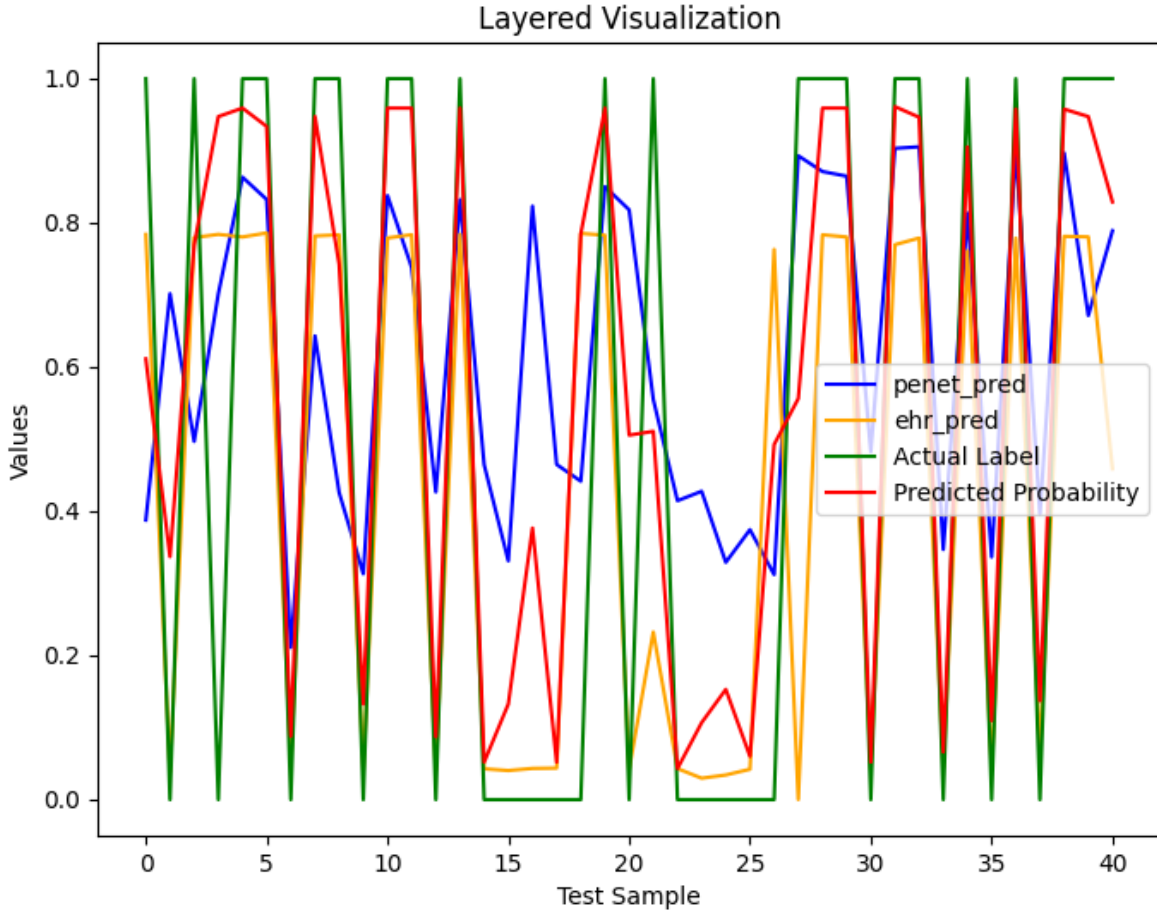


Fig. 3. Prediction trend of different validation samples using different models in comparison to their actual labels

TABLE II
PERFORMANCE COMPARISON OF MODELS

Model	Accuracy	Precision	Recall	F1 Score	AUC
Image-Only Models					
PENet	0.77	0.81	0.73	0.77	0.84
EHR-Only Models					
TabNet	0.85	0.84	0.90	0.86	0.87
ElasticNet	0.83	0.90	0.80	0.84	0.92
Multimodal Models					
XGBoost Fusion	0.88	0.83	0.94	0.86	0.85
TabNet Fusion	0.88	0.83	0.94	0.86	0.85
CatBoost Fusion	0.93	0.87	0.95	0.90	0.96

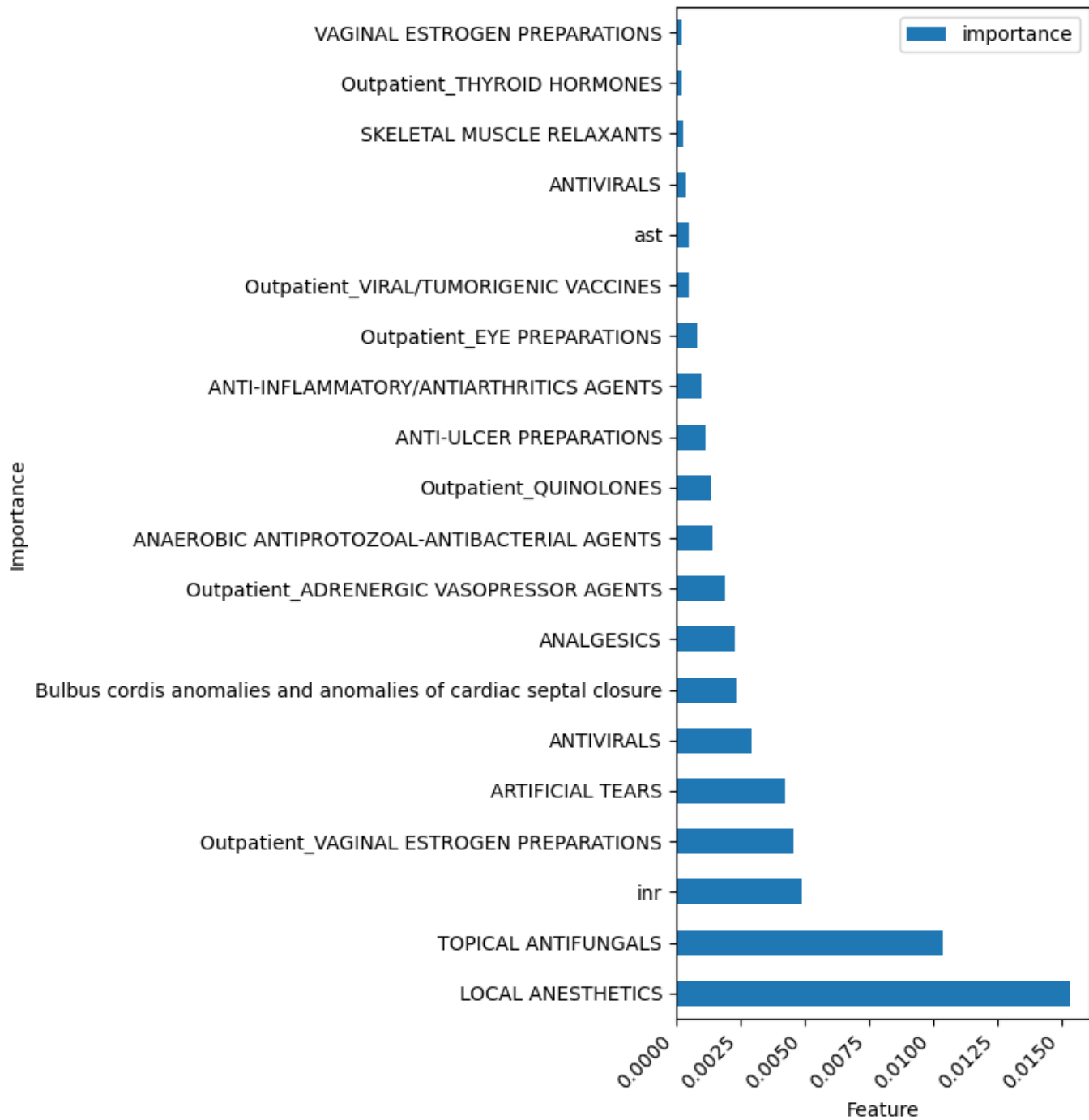


Fig. 4. Feature Importance other than Diseases of Pulmonary Circulation

play a significant role in predicting disease outcomes, emphasizing the importance of incorporating medication history and treatment plans into predictive models for enhanced accuracy.

Additionally, the inclusion of INR (International Normalized Ratio) as a feature with a feature importance score of 0.005 underscores the relevance of coagulation status in disease prediction, particularly in conditions where blood clotting disorders may influence clinical outcomes. Overall, our feature importance analysis provides valuable insights into the factors driving predictive performance in our EHR-based model, offering clinicians and researchers a deeper understanding of the underlying mechanisms influencing disease prognosis and treatment outcomes.

The visualization presented in the result section illustrates the alignment between the model's predictions and the actual labels

using the validation set. The plot showcases the values of the PENet predictions and TabNet prediction features alongside the corresponding actual labels and predicted probabilities. This graphical representation offers a comprehensive overview of the model's performance, allowing for an intuitive understanding of its predictive capabilities. By visually comparing the predicted probabilities with the actual labels, one can assess the model's accuracy and its ability to correctly classify instances.

V. DISCUSSION

A. Limitations

- 1) **Availability of Multimodal Dataset:** One of the primary limitations of this study is the scarcity of publicly available multimodal datasets. While multimodal datasets offer a rich source of information and potential insights, they are often challenging to obtain due to privacy concerns, data access restrictions, and the need for integration across multiple data sources.
- 2) **Consistency in Features:** Another limitation is the lack of consistency in features across modalities. Different modalities may have varying levels of completeness and consistency, leading to challenges in feature engineering and model training. This inconsistency can introduce bias and affect the performance of multimodal models.
- 3) **Ethical and Regulatory Considerations:** Working with multimodal healthcare data involves ethical and regulatory considerations, including privacy protection, informed consent, and compliance with data protection regulations such as GDPR and HIPAA. These considerations may limit the accessibility and usability of healthcare datasets for research purposes.

B. Further Discussion and Possible Improvements

- 1) **Data Collection and Integration:** Efforts should be made to collect and integrate multimodal data from diverse sources, including healthcare institutions, research organizations, and data-sharing initiatives. Collaborations and partnerships can facilitate access to a broader range of data types, enabling more comprehensive analyses and model development.
- 2) **Feature Engineering and Harmonization:** Improving the consistency and quality of features across modalities is essential for the development of robust multimodal models. Standardization protocols, such as ontologies and data dictionaries, can be employed to ensure uniformity in data representation and semantics. Advanced feature engineering techniques and data preprocessing methods can help address discrepancies and enhance the compatibility of multimodal features.
- 3) **Model Robustness and Generalization:** Enhancing the robustness and generalization of multimodal models is crucial for their real-world applicability. Techniques such as transfer learning, domain adaptation, and model ensembling can leverage knowledge from related tasks or domains to improve performance on target tasks. Additionally, model interpretability methods can provide insights into the decision-making process of multimodal models, enhancing trust and transparency.
- 4) **Ethical and Regulatory Compliance:** Ethical and regulatory considerations must be carefully addressed when working with multimodal healthcare data. Privacy protection, informed consent, data anonymization, and compliance with data protection regulations are critical aspects that require attention to ensure the responsible and ethical use of sensitive patient information.

In conclusion, while multimodal datasets hold great promise for advancing healthcare research and improving patient outcomes, addressing the challenges associated with data availability, consistency, and ethical considerations is essential for realizing their full potential. By overcoming these limitations and implementing appropriate strategies for improvement, multimodal models can play a significant role in personalized medicine, disease diagnosis, and treatment planning.

VI. CONCLUSION

In this study, we explored various machine learning models for the task of PE detection. We experimented with PENet, TabNet, XGBoost, CatBoost, etc., leveraging both individual modalities and fusion techniques to improve predictive performance. Through comprehensive experimentation and evaluation, several key findings have emerged.

We conducted a thorough performance comparison of different models, evaluating metrics such as accuracy, precision, recall, F1-score, and AUC-ROC. Our results indicate that the decision-level fusion (Late fusion) model using the CatBoost meta-learner and the PENet+TabNet individual models performs the best, achieving a high accuracy of 0.93 on the test set II.

This finding suggests that leveraging both PENet and TabNet models in combination with decision-level fusion can significantly enhance the accuracy of PE detection, offering potential benefits for clinical diagnosis and patient care.

However, it's essential to acknowledge the limitations of our study. We faced challenges related to the availability of multimodal datasets and ensuring consistency in the features across different modalities. Additionally, while our best-performing model demonstrates promising results, further research is needed to address model interpretability and generalizability issues.

In conclusion, our study highlights the efficacy of machine learning models and fusion techniques for PE detection. Future research directions may involve refining existing models, incorporating additional features or modalities, and conducting prospective clinical validation studies to assess the real-world performance of these models.

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